

The new release includes frame pointers enabled by default across all 64-bit architectures. “Frame pointers allow more complete CPU profiling and off-CPU profiling. The performance wins that these can provide far outweigh the comparatively tiny loss in performance. Ubuntu enabling frame pointers by default will be a huge win for performance engineering and the



default developer experience,” said Brendan Gregg, computer performance expert and fellow at Intel. Tracing with *bpfttrace* is now standard in Ubuntu 24.04 LTS, alongside pre-existing profiling tools to provide site reliability engineers with immediate access to essential resources.

The latest release also comes with new management tools for Ubuntu Desktop and WSL. For the first time in an LTS, Ubuntu Desktop uses the same installer technology as Ubuntu Server. This means that desktop administrators can now use image customisation

tools like *autoinstall* and *cloud-init* to create tailored experiences for their developers. The user interface has also received a makeover, with a modern design built in Flutter.

Local AI code completion possible with JetBrains 2024.1

JetBrains has rolled out the 2024.1 updates for its IDEs, introducing significant new features, with the spotlight on full-line code autocompletion driven by local AI models.

This latest addition, full-line code completion, guarantees that code suggestions are generated entirely on the user’s device, reducing latency and offering a smooth offline experience.

“We’ve developed models that run directly on your device, and the IDE verifies each suggestion,” explained Mikhail Kostyukov, product manager in JetBrains Machine Learning Team. “This approach delivers the best user experience and ensures

that the generated code never contains non-existent variables or methods.”

The full-line code completion feature is included with existing paid JetBrains IDE subscriptions for a wide range of languages, including Java, Kotlin, Python, JavaScript, TypeScript, CSS, PHP, Go, and Ruby. Support for additional languages like C#, Rust, and C++ will be added in the coming months across relevant IDEs such as Rider, RustRover, and CLion Nova.



In addition to local AI capabilities, the 2024.1 releases introduce an enhanced terminal with visual improvements, command completion, improved navigation of nested commands, and easier access to history. Remote development also receives performance and stability enhancements, along with expanded technology support.

Updates specific to individual IDEs include Java 22 support in IntelliJ IDEA, a new K2 mode in Kotlin for enhanced analysis, VCS command execution using the project SDK in RubyMine, the exit of data flow analysis from beta in GoLand, Jupyter notebook improvements in PyCharm, and Symfony AssetMapper support in PhpStorm.

The use of local AI models is particularly beneficial for industries with stringent data privacy regulations, such as healthcare and finance, as well as for developers operating in protected infrastructure or with limited internet access.

To ensure privacy, JetBrains emphasises that its AI solutions are not trained on any customer code. The 2024.1 releases are now available for download. Additionally, JetBrains’ AI Assistant offers enhancements like test generation, cloud completion, custom commit prompts, and in-editor code block generation.

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